

Joon-Hwi Kim

Ph.D., Physics

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Research Subjects

General Relativity; Scattering Amplitudes; Twistor Theory; Differential Geometry; Generalized Symmetry;

Education

Ph.D. in Physics, Caltech (Pasadena, California, USA)	09/2021-06/2026
Advisor: Clifford Cheung	
B.S. in Physics, Seoul National University (Seoul, Korea)	02/2017-02/2021
Advisor: Sangmin Lee	
Seoul Science High School for the Gifted (Seoul, Korea)	02/2014-02/2017

Publications

[2405.17056]	Massive twistor worldline in electromagnetic fields JHK , J.-W. Kim, S. Lee <i>J. High Energy Phys.</i> 8 (2024) 080	30
[2410.22988]	Classical eikonal from Magnus expansion JHK , J.-W. Kim, S. Kim, S. Lee <i>J. High Energy Phys.</i> 1 (2025) 111	26
[2403.01837]	Generalized symmetry in dynamical gravity C. Cheung, M. Derda, JHK , V. Nevoa, I. Rothstein, N. Shah <i>J. High Energy Phys.</i> 10 (2024) 007	23
[2405.09518]	Single Kerr-Schild metric for Taub-NUT instanton JHK <i>Physical Review D</i> 111 (2025) 2, L021703	20
[2301.06203]	Symplectic perturbation theory in massive twistor space: a zig-zag theory of massive spinning particles JHK , S. Lee	18
[2309.11886]	Asymptotic spinspace-time JHK <i>Physical Review D</i> 111 (2025) 10, 105011	16
[2412.19611]	Newman-Janis algorithm from Taub-NUT instantons JHK <i>Physical Review Letters</i> 136 (2026) 23, 231401	15
[2102.07063]	The relativistic spherical top as a massive twistor JHK , J.-W. Kim, S. Lee <i>J. Phys. A</i> 54 (2021) 33, 335203	14
[2511.07387]	Manifest symplecticity in classical scattering JHK <i>J. High Energy Phys.</i> 5 (2026) 287	9

[2512.23697]	Note on the Kerr spinning-particle equations of motion JHK	7
[2509.23600]	Geodesic deviation to all orders via a tangent bundle formalism JHK <i>General Relativity and Gravitation</i> 58 (2026) 7 75	6
[2509.06058]	Worldline formalism in phase space JHK	5
[2602.19495]	The Kerr two-twistor particle JHK	4
[2603.07537]	The Kerr-Newman two-twistor particle JHK	3
[2512.23100]	Phase space formulation of S-matrix JHK	3
[2603.27353]	Universality in relativistic spinning particle models JHK , S. Lee (<i>Under review, J. High Energy Phys.</i>)	2
[2510.23080]	Double copy and the double Poisson bracket	2
[2603.21934]	Covariant symplectic geometry of classical particles JHK	1
[2605.25473]	The diagrammar of quantum Magnusian L. Guo, JHK , J.-W. Kim, S. Kim, S. Lee, J.-R. Li	0
[2512.21210]	Twisted Feynman integrals: from generating functions to spin-resummed post-Minkowskian dynamics JHK , J.-W. Kim, J. Lim (<i>Under review, J. High Energy Phys.</i>)	0

Grants & Awards

Research Grant, Ilju Academy of Science of Culture (\$120,000)	08/2021-07/2025
Presidential Science Scholarship, Korea Student Aid Foundation (Full tuition)	03/2017-02/2021
Korea Talent Award (Korea Foundation for Advancement of Science & Technology, 50 individuals/nation)	11/2016
Gold Prize in the 5th Hanwha Science Challenge (₩20,000,000, Hanwha Co., Ltd.)	08/2015
Gold Medal in the 46th International Physics Olympiad (Tata Institute of Fundamental Research)	07/2015
Hansung Nobel Scholarship for the Gifted (Hansung Scholarship Foundation)	03/2015-02/2016

Invited Talks

Worldline Formalism in Phase Space Worldline Seminars	10/28/2025
Building Black Holes from Taub-NUT Instantons Self-Force Group Seminar Series, University of Southampton	02/05/2025
Newman-Janis Shift and Beyond Séminaire Amplitudes et Gravitation sur l'Yvette, IHÉS/IPhT (Bures-sur-Yvette, France)	01/15/2025
One-Form Symmetry in Dynamical Gravity Theoretical Elementary Particle (TEP) Physics Seminar, UCLA (Los Angeles, California, USA)	04/18/2024
Twistor Particle Programme Rebooted: A Zig-Zag Theory of Massive Spinning Particles QFT-Relativity Seminars, Oxford University	11/17/2023

Seminars and Colloquia

Spinning Black Hole Amplitudes from Complexified Equivalence Principle Amplitudes 2024 Gong Show, Institute for Advanced Study (New Jersey, USA)	06/12/2024
Spinspacetime: Unification of Spacetime and Spin into a Complex Geometry 39th Pacific Coast Gravity Meeting, Caltech (Pasadena, California, USA)	03/31/2023
Symplectic Perturbation Theory California Amplitudes Meeting 2023, UCSD (San Diego, California, USA)	03/12/2023

Conferences and Schools

Twistors in Geometry & Physics (Isaac Newton Institute for Mathematical Sciences, Cambridge, UK; Invitation by Lionel Mason)	09/2024
Advanced School on Quantum Spacetime (University of Warsaw, Warsaw, Poland)	09/2024
Mathematical Structures in Scattering Amplitudes (Galileo Galilei Institute for Theoretical Physics, Italy; Invitation by Lionel Mason)	08/2024
IHÉS Summer School – Symmetries and Anomalies: a Modern Take (Institut des Hautes Études Scientifiques, Bures-sur-Yvette, France)	06-07/2024
Amplitudes 2024 Summer School (Institute of Advanced Study, Princeton, New Jersey, USA)	06/2024
Advanced School on Asymptotically Flat Spacetime and Celestial Holography (University of Warsaw, Warsaw, Poland)	08-09/2023
KIAS-SNU Winter Camp (Korea Institute of Advanced Study, Seoul, Korea)	12/2019
Quantum Fields and Strings on Curved Geometry (Korea Institute of Advanced Study, Seoul, Korea)	07-08/2019
Summer School on Numerical Relativity (Asian-Pacific Center of Theoretical Physics, Seoul, Korea)	06/2019

Teaching & Service

Teaching Assistant

- ❑ Caltech (2022-2026)
 - Theoretical Cosmology and Astroparticle Physics, Ph 235b Spring 25-26
 - Mathematical Methods of Physics, Ph 129b Winter 25-26
 - General Relativity, Ph 236a Fall 25-26
 - Topics in Classical Physics (electrodynamics and relativity), Ph 106c Spring 24-25
 - Topics in Classical Physics (electromagnetism), Ph 106b Winter 24-25
 - Topics in Classical Physics (mechanics), Ph 106a Fall 24-25
 - Quantum Mechanics, Ph 125c Spring 23-24
 - General Relativity, Ph 236b, prof. Saul Teukolsky Winter 23-24
 - General Relativity, Ph 236a Fall 23-24
 - Candidacy Exam Fitness, Ph 201 Spring 22-23
 - Quantum Mechanics, Ph 2b Winter 22-23
 - Classical Mechanics, Ph 1a Fall 22-23
- ❑ Korean Physics Society (2017-2019)
 - Theoretical and Experimental Problem Solving for International Physics Olympiad

Volunteering

- ❑ Caltech (2022-2025)
 - Future Ignited Conference 09/2022, 2023

Lecturer

- Physics and the Imagination 03-12/2017
 - 10 lectures on mathematical physics (quaternions, spinors and Clifford algebra, matrix representations, differential forms, de Rham cohomology); Refs. Schutz (1980), Nakahara (2003), Doran (2003)
- Escher's Studio 03-12/2017
 - Interdisciplinary program that explored the intersections between mathematics (group theory, homology, complex analysis), physics (symmetry, representation theory, gauge theory), computer science (Javascript programming, generative art), visual art (tessellations, planar compositions, illusions, impossible figures, perspectives, drawing on nonlinear grids), and music (musical tessellations in chord space, Shepard tone).

Referee Role

- European Physical Journal C (2025), International Journal of Geometric Methods in Modern Physics (2026), International Journal of Theoretical Physics (2026)

Coursework

Caltech

- Introduction to String Theory, Ph 250b Spring 24-25
- Introduction to String Theory, Ph 250a Winter 24-25
- Introduction to Knot Theory and Quantum Topology, Ma 146b Winter 23-24
- Topics in Mathematical Physics, Ma 148a Fall 22-23
- Advanced Mathematical Methods of Physics, Ph 229b Spring 21-22
- Elementary Particle Theory, Ph 230b Spring 21-22
- Riemannian Geometry, Ma 157b Winter 21-22
- Relativity, Ph 236b Winter 21-22
- Physics Seminar, Ph 242b Winter 21-22
- Mathematical Methods of Physics, Ph 129a Fall 21-22
- Relativistic Quantum Field Theory, Ph 205a Fall 21-22
- Advanced Mathematical Methods of Physics, Ph 229a Fall 21-22
- Elementary Particle Theory, Ph 230a Fall 21-22
- Introduction to Topological Field Theory, Ph 232 Fall 21-22
- Physics Seminar, Ph 242a Fall 21-22

Seoul National University

- Particle Physics (Graduate) 2020
- Properties of Solids 2020
- Classical Mechanics (Graduate) 2020
- Quantum Mechanics I (Graduate) 2020
- Quantum Fields and Strings on Curved Spacetime (Graduate, KIAS SITP) 2019
- Quantum Field Theory II (Graduate) 2019
- Quantum Field Theory I (Graduate) 2019
- Elementary Nuclear and Particle Physics (Graduate) 2019
- Theory of Lie Groups (Graduate) 2019
- Computational Linguistics 2018
- Artificial Intelligence for Physicists 2018
- Fluid Dynamics 2017
- Relativity and Spacetime 2017